

Assignment

① Naive Bayes Conditional Probability เหตุการณ์ / ความน่าจะเป็น ที่เกี่ยวข้อง

$$P(B) \times P(A|B) = \frac{P(A \cap B)}{P(B)}$$

เหตุการณ์ B แล้วจึงหาค่าของเหตุการณ์ A ตามสมการ

join probability ความน่าจะเป็นที่ จะเกิดเหตุการณ์ A และ B ร่วมกัน

วิธีทำ คำนวณค่าความน่าจะเป็นของค่าที่ 110 หรือ 3 ปี



} dataset = 9 Income > 50k = 3/9 = 0.33

Income <= 50k = 6/9 = 0.67

attribute	Income > 50k (3)	Income <= 50k (6)
work - Self	2/3 = 0.67	1/6 = 0.17
- Private	1/3 = 0.33	5/6 = 0.83
ed - Bachelors	1/3 = 0.33	2/6 = 0.33
- HS-grad	1/3 = 0.33	3/6 = 0.5
- Masters	1/3 = 0.33	1/6 = 0.17
marital - Never	2/3 = 0.67	2/6 = 0.33
- Married	1/3 = 0.33	4/6 = 0.67
occu - Prof	2/3 = 0.67	1/6 = 0.17
- Exec	1/3 = 0.33	3/6 = 0.5
- Handlers	0/3 = 0	2/6 = 0.33
relation - Not	2/3 = 0.67	2/6 = 0.33
- Husband	1/3 = 0.33	2/6 = 0.33
- Wife	0/3 = 0	2/6 = 0.33
sex = Male	2/3 = 0.67	3/6 = 0.5
- Female	1/3 = 0.33	3/6 = 0.5

Testing data

Work class : Private

Education : Bachelors

Marital-status : Married - civ - spouse

Occupation : Exec - managerial

Relationship : Husband

Sex : Male

Assumption Income $> 50k$

$$\begin{aligned}P(\text{Income} > 50k | A) &= P(\text{work} = \text{Private} | \text{Income} > 50k) \times P(\text{ed} = B | \text{Income} > 50k) \times \\&\quad P(\text{marital} = \text{Married} | \text{Income} > 50k) \times P(\text{occ} = \text{Exec} | \text{Income} > 50k) \times \\&\quad P(\text{rela} = \text{Husband} | \text{Income} > 50k) \times P(\text{sex} = \text{Male} | \text{Income} > 50k) \times \\&\quad P(\text{Income} > 50k) \\&= 0.33 \times 0.33 \times 0.33 \times 0.33 \times 0.33 \times 0.67 \times 0.33 \\&= 0.001\end{aligned}$$

Assumption Income $\leq 50k$

$$\begin{aligned}P(\text{Income} \leq 50k | A) &= P(\text{work} = \text{Private} | \text{Income} \leq 50k) \times P(\text{ed} = B | \text{Income} \leq 50k) \times \\&\quad P(\text{marital} = \text{Married} | \text{Income} \leq 50k) \times P(\text{occ} = \text{Exec} | \text{Income} \leq 50k) \times \\&\quad P(\text{rela} = \text{Husband} | \text{Income} \leq 50k) \times P(\text{sex} = \text{Male} | \text{Income} \leq 50k) \times \\&\quad P(\text{Income} \leq 50k) \\&= 0.83 \times 0.33 \times 0.67 \times 0.5 \times 0.33 \times 0.5 \times 0.67 \\&= 0.010\end{aligned}$$

$\therefore$ on testing data probabilities $\leq 50k$

② KNN ($n = 5$)

Mass Width Height Skin Color Animal Type
...

$$d(x, B_j) = \sqrt{\sum (p_k - q_k)^2}$$

$$d(x, B_1) = \sqrt{(157 - 142)^2 + (7.3 - 7.3)^2 + (7.2 - 7.2)^2 + (0.88 - 0.88)^2} = 15.00$$

$$d(x, B_2) = \sqrt{(157 - 355)^2 + (7.3 - 9)^2 + (7.2 - 9.5)^2 + (0.88 - 0.75)^2} = 178.02$$

$$d(x, B_3) = \sqrt{(157 - 165)^2 + (7.3 - 7)^2 + (7.2 - 7.5)^2 + (0.88 - 0.82)^2} = 8.01$$

$$d(x, B_4) = \sqrt{(157 - 185)^2 + (7.3 - 8)^2 + (7.2 - 6.8)^2 + (0.88 - 0.59)^2} = 28.01$$

$$d(x, B_5) = \sqrt{(157 - 87)^2 + (7.3 - 6.2)^2 + (7.2 - 4.7)^2 + (0.88 - 0.61)^2} = 70.05$$

$$d(x, B_6) = \sqrt{(157 - 192)^2 + (7.3 - 8.4)^2 + (7.2 - 7.3)^2 + (0.88 - 0.54)^2} = 35.02$$

$$d(x, B_7) = \sqrt{(157 - 153)^2 + (7.3 - 7.1)^2 + (7.2 - 7.9)^2 + (0.88 - 0.75)^2} = 4.07$$

$$d(x, B_8) = \sqrt{(157 - 85)^2 + (7.3 - 5.8)^2 + (7.2 - 4.5)^2 + (0.88 - 0.78)^2} = 72.07$$

$$K_1 = 7 \rightarrow \text{cow}$$

$$K_2 = 7, 3 \rightarrow \text{cow, cow}$$

$$K_3 = 7, 3, 1 \rightarrow \text{cow, cow, cow}$$

$$K_4 = 7, 3, 1, 4 \rightarrow \text{cow, cow, cow, apple}$$

$$K_5 = 7, 3, 1, 4, 6 \rightarrow \text{cow, cow, cow, apple, Elephant}$$

$\therefore$ Animal Type (x) = cow